

USER MANUAL

Circuit Lab 2.0

A breadboard, a PCB, and twenty real projects —
from your first LED to a finished board you can
export.

17 real components

20 guided projects

Live circuit simulation

PCB export

Why Circuit Lab

This isn't a slideshow or a quiz. It's a live breadboard that actually simulates real circuit physics — wires you drag genuinely carry current, LEDs genuinely light up (and burn out if you overdrive them), and capacitors genuinely charge and discharge in real time.

It starts from true zero — what a battery, resistor, and switch even do — and walks you through twenty hands-on projects covering series and parallel circuits, potentiometers, capacitors and their different types, 555 timers, transistors, MOSFETs, relays, diodes, AC power and bridge rectifiers, inductors, photoresistors, solar power, and finishes with real PCB design, complete with vias and an exportable board file.

Every project has a concrete goal, a hint if you get stuck, and a **Cheat** button that shows a working solution as a proper schematic if you want to peek — plus a **Schematic** button that generates a clean diagram from whatever you actually built, which is a genuinely useful skill on its own.

It's satisfying in the way real tinkering is satisfying: you wire something wrong, something pops or refuses to light, you get a plain-English explanation of why, and you fix it — except here mistakes are free and instant instead of costing you a part and a trip to the store.

New to electronics entirely? There's also a simpler **Circuit Lab 1.0**, a 7-project version covering just the fundamentals — batteries, resistors, LEDs, switches, capacitors, and a 555 timer — if you'd rather start gentler before jumping into 2.0's deeper end.

What you'll need

Just a browser. Open the file, no installs, no accounts. Everything — audio included — runs locally.

Getting Started

The parts bin (left)

Click a part to add it to the board. Parts you haven't unlocked yet for the current project show up greyed out — they unlock automatically as you reach the projects that use them.

The board (center)

Move a part	Click and drag its body.
Wire two pins	Click and drag from one connector dot to another.
Delete a wire	Click directly on it — it highlights red on hover.
Route around things	While drawing a wire, click empty board space to drop a bend point and keep going.
Change a value	Tap most parts to cycle their setting (resistance, battery type, capacitor type...). Parts with a slider (rheostat, photoresistor, capacitor value) drag live.
Remove a part	Hover it and click the small × that appears in its corner.

Header tools

Tilt view	A cosmetic 3D angle on the board — click again to return flat.
PCB mode	Switches the board's skin to a green PCB with copper traces (unlocks with the PCB project).
Clear board	Removes everything from the current project so you can start over.
Save	Signs and saves your current circuit — asks for your initials the first time.
Schematic	Draws a proper schematic diagram from whatever you've actually wired.
Cheat	Shows a working schematic solution for the current project, if you're stuck.
Export Design	Appears in the PCB project — downloads your board as an SVG plus a parts/connection list.
Sound	Mutes or unmutes all audio — connection sounds, buzzers, the ambient hum of current flowing.

The level track (bottom)

Click any unlocked circle to jump straight to that project. Your progress and each project's board are saved automatically — leave and come back anytime.

The Parts Bin

All seventeen components, and what each one actually does.



Battery

Powers the circuit. Tap to cycle through six real chemistries — AA, 9V, Li-Po, lead-acid, solar, even AC.



LED

Lights up when current flows the correct direction. Burns out if overdriven — always give it its own resistor.



Rheostat

A variable resistor. Drag its slider to change resistance live, while the circuit is running.



555 Timer

A classic oscillator chip. Blinks an LED on its own — no capacitor math required.



Buzzer

Makes a real audible tone (through your speakers) whenever current flows through it.



Photoresistor

A light-dependent resistor. Its resistance changes with light — drag its slider to simulate day and night.



Bridge Rectifier

Four diodes in one part. Turns alternating current (AC) into pulsing direct current (DC).



MOSFET

A voltage-controlled switch. Needs almost no current at its gate to switch a much larger load.



Motor

Spins faster the more current flows through it — a simple, visible way to see current at work.



Resistor

Limits current through a branch. Tap to cycle preset resistance values.



Switch

A toggle — click once to close the circuit, click again to open it.



Capacitor

Stores electrical charge. Five real construction types — some are polarized and can pop if wired backwards.



Push Button

A momentary switch — only closes while you are physically holding it down.



Transistor

A current-controlled switch. A small current into its base switches a much larger current on or off.



Diode

Lets current flow in one direction only — the basic building block of rectification.



Inductor

Resists sudden changes in current, producing a gradual ramp-up instead of an instant snap-on.



Relay

An electromechanical switch. Lets a small, safe control circuit switch a completely separate circuit.

The 20 Projects

Goals only here — no wiring spoilers. If you get stuck, the in-project hint or the Cheat button will get you unstuck.

Act I — The Basics

Series and parallel circuits, resistors, and your first working LED loop.

#	Project	Goal
1	Light It Up	Get the LED lit while the switch is closed.
2	Two In Series	Light two LEDs wired in series from one battery.
3	Parallel Power	Light two LEDs wired in parallel, each with its own resistor.
4	The Dimmer	Get the LED brightness above 70% using the rheostat.

Act II — Timing & Boards

Capacitors, the 555 timer chip, and your first real PCB layout.

#	Project	Goal
5	The Afterglow	Open the switch and watch the LED glow for a moment before fading out, powered only by the capacitor.
6	The Timer Chip	Get an LED blinking from the 555 timer output.
7	PCB Designer	Light at least one LED on the PCB board.

Act III — Switches That Aren't Switches

Push buttons, buzzers, transistors, and light-activated circuits.

#	Project	Goal
8	The Doorbell	Get the buzzer sounding while you hold the button down.
9	The Amplifier	Press the button to switch the LED on, then release it and watch the LED turn off.
10	The Night Light	Drag the photoresistor toward the moon until the LED switches on by itself.
11	The Alarm	Make the buzzer sound on its own when you drag the photoresistor toward the dark.

Act IV — Power Electronics

Diodes, AC rectification, capacitor types, and inductors.

#	Project	Goal
12	The Rectifier Diode	Get the LED lit through the diode.
13	AC & The Bridge	Get an LED lit using power from an AC source, rectified through the bridge.

#	Project	Goal
14	Smooth Operator	Keep the LED continuously lit, with no flicker, for 3 seconds straight.
15	Choosing Your Capacitor	Smooth the same circuit using a capacitor actually labeled Electrolytic or Supercapacitor.
16	The Inductor's Delay	Close the switch and watch the LED ease up to full brightness instead of switching on instantly.

Act V — Going Further

MOSFETs, relays, real-world power sources, and a finished PCB export.

#	Project	Goal
17	Transistor vs MOSFET	Switch one LED on with the transistor, and a separate LED on with the MOSFET.
18	The Relay & Flyback Diode	Close the control switch and get the relay to switch on a separate LED circuit.
19	Power Sources	Power the LED using the Solar Panel battery type, with its slider up enough to light it.
20	PCB Vias & Export	Route at least one trace through a via, and light the LED.

When Something Pops

Parts can genuinely burn out here, just like in real life — it's not a bug, it's the simulator teaching you the same lesson a real breadboard would, without costing you an actual part. When it happens, a message explains exactly why. The most common causes:

An LED with no resistor of its own. Even if there's already a resistor somewhere else in the circuit, every single LED needs its own resistor in series. It's an easy one to miss in circuits that already have a resistor doing another job nearby (see Project 10).

A polarized capacitor wired backwards. Electrolytic, tantalum, and supercapacitors care which way round they go — the + lead needs to be on the higher-voltage side. Ceramic and film capacitors don't care; they're safe either way.

Too much current, generally. Diodes, buzzers, bridge rectifiers, and motors can all overheat if there's nothing limiting the current — a short circuit, or a load wired straight across a battery with no resistor at all.

A burnt part gets a visible scorch mark and a small  above it so you don't lose track of it. Delete it and drop in a fresh one — the rest of your wiring stays put.

Other things worth knowing

- **LEDs only work one way round.** The flat side of the dome marks the cathode (–); if nothing lights up, try flipping it.
- **The power gauge** (top of the screen) shows live voltage and current the moment your loop is actually complete — a quick way to check your circuit is really connected.
- **The Schematic button** is a great sanity check whenever a circuit isn't behaving the way you expect — seeing your own wiring redrawn as a clean diagram often reveals the mistake immediately.
- **Short circuits** flash the board and buzz an alarm — that's the simulator protecting you from a wiring mistake, exactly like a real fuse would.